

HYBRIDISATION

JEE Advanced Preparation - Chemistry

Chemical Bonding & Molecular Structure

Topic Overview

Hybridisation is the concept of mixing of atomic orbitals of similar energies to form a new set of equivalent orbitals called **hybrid orbitals**. It explains the geometry, bond angles, and shapes of molecules that cannot be predicted by simple atomic orbital theory.

This module covers: sp , sp^2 , sp^3 , sp^3d , sp^3d^2 , sp^3d^3 hybridisation; lone pair effects; worked examples; and JEE Advanced MCQs.

1. THEORY OF HYBRIDISATION

1.1 Definition and Concept

Hybridisation is the process of mixing of two or more atomic orbitals of an atom having nearly the same energy to produce an equal number of identical new orbitals called **hybrid orbitals**. These hybrid orbitals have the same energy, identical shape and are equivalent in all respects.

Key Characteristics of Hybrid Orbitals:

- Hybrid orbitals have one large lobe and one small lobe.
- Hybrid orbitals form stronger bonds than pure s or p orbitals due to greater directional overlap.
- Number of hybrid orbitals formed = Number of atomic orbitals mixed.
- Lone pairs occupy hybrid orbitals just like bond pairs.

1.2 Formula for Hybridisation Number

$H = (V + M - C + A) / 2$ where: V = number of valence electrons of central atom M = number of monovalent atoms attached C = charge on the cation (if ionic species) A = charge on the anion (if ionic species)
H = 2 → sp | H = 3 → sp² | H = 4 → sp³ H = 5 → sp³d | H = 6 → sp³d² | H = 7 → sp³d³

1.3 Types of Hybridisation

sp Hybridisation (Linear, 180°)

One s orbital mixes with one p orbital to form 2 equivalent sp hybrid orbitals. The two sp orbitals orient themselves at 180° to minimise repulsion. The remaining two p orbitals are perpendicular to the molecular axis and are available for pi bonding.

Examples: BeCl₂ (no lone pairs, linear), CO₂ (O=C=O, 2 double bonds), C₂H₂ (acetylene, triple bond), HCN (H-C≡N)

sp² Hybridisation (Trigonal Planar, 120°)

One s orbital mixes with two p orbitals to form 3 equivalent sp² hybrid orbitals arranged in a plane at 120°. The remaining unhybridised p orbital is perpendicular to this plane and participates in pi bonding.

Examples: BF₃ (no lone pairs), SO₃, C₂H₄ (ethylene), NO₃⁻ (carbonate-like resonance)

sp³ Hybridisation (Tetrahedral, 109.5°)

One s orbital mixes with three p orbitals to form 4 equivalent sp³ hybrid orbitals directed towards the corners of a regular tetrahedron at 109.5°. Lone pairs occupy sp³ orbitals and compress bond angles.

Examples: CH₄ (109.5°), NH₃ (1 lone pair, 107°), H₂O (2 lone pairs, 104.5°), CCl₄, NF₃

sp³d Hybridisation (Trigonal Bipyramidal, 90°/120°)

One s, three p, and one d orbital combine to form 5 sp³d hybrid orbitals. Three equatorial orbitals are at 120° in a plane; two axial orbitals are at 90° to the equatorial plane. Lone pairs preferentially occupy equatorial positions to minimize repulsion.

Examples: PCl₅ (no lone pairs), SF₄ (1 lone pair, see-saw), ClF₃ (2 lone pairs, T-shaped), XeF₂ (3 lone pairs, linear)

sp³d² Hybridisation (Octahedral, 90°)

One s, three p, and two d orbitals combine to form 6 sp³d² hybrid orbitals pointing to the six corners of an octahedron. All bond angles are 90°.

Examples: SF₆, PCl₆⁻, [Co(NH₃)₆]³⁺

sp³d³ Hybridisation (Pentagonal Bipyramidal)

Seven hybrid orbitals formed by mixing one s, three p, and three d orbitals. Five equatorial orbitals at 72° and two axial orbitals at 90° to the equatorial plane.

Example: IF₇

1.4 Effect of Lone Pairs

Lone pairs occupy more space than bond pairs because they are attracted by only one nucleus (instead of two). This causes greater repulsion and reduces bond angles. The **geometry** describes actual 3D arrangement of atoms only (excluding lone pairs), while **hybridisation** includes lone pairs.

Lone Pair Effects in sp³ Molecules:

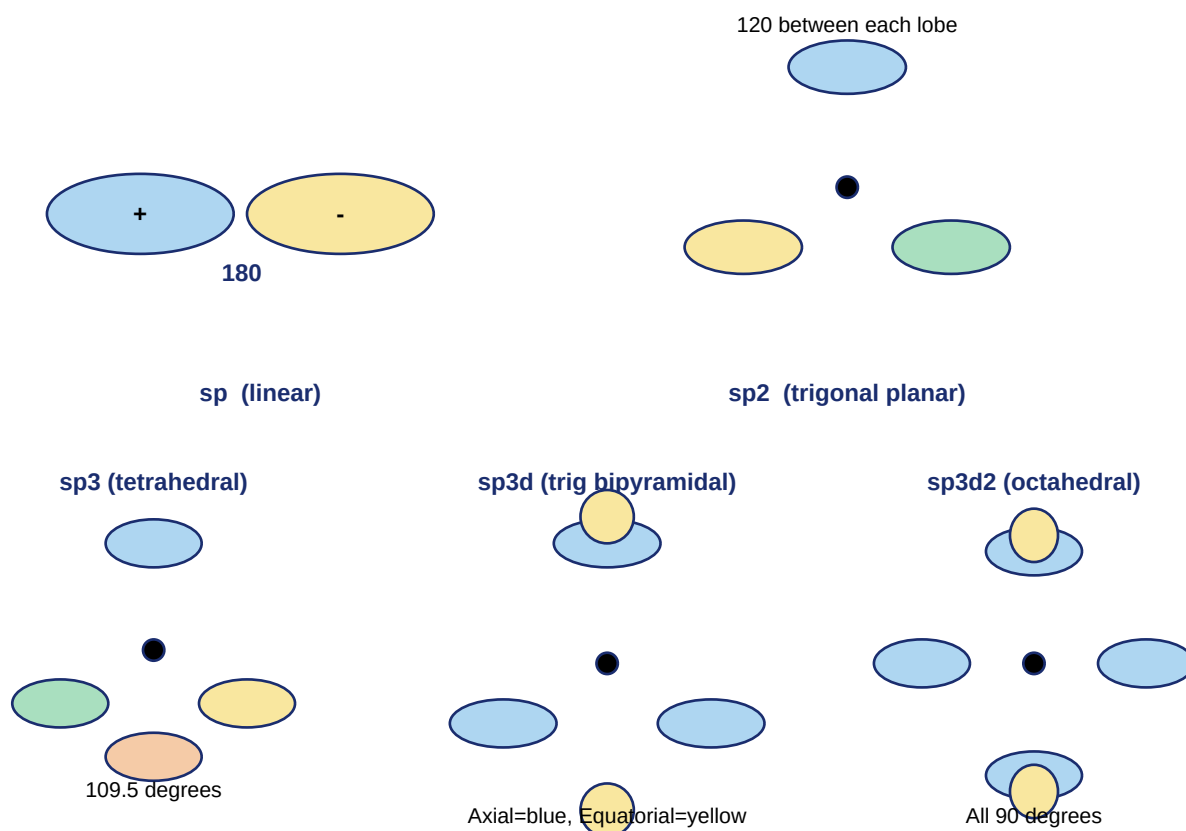
CH₄: 0 lone pairs, tetrahedral geometry, 109.5°

NH₃: 1 lone pair, trigonal pyramidal geometry, 107°

H₂O: 2 lone pairs, bent/V-shaped geometry, 104.5°

1.5 Orbital Diagrams

The diagrams below show the arrangement of hybrid orbital lobes (large positive lobe and small negative lobe):



1.6 Summary Table

Hybridisation	Geometry	Bond Angle	Examples
sp	Linear	180 deg	BeCl ₂ , CO ₂ , C ₂ H ₂ , HCN
sp ²	Trigonal Planar	120 deg	BF ₃ , SO ₃ , C ₂ H ₄ , NO ₃ ⁻
sp ³	Tetrahedral	109.5 deg	CH ₄ , CCl ₄ , SiH ₄
sp ³	Trigonal Pyramidal	107 deg	NH ₃ , PCl ₃ , NF ₃
sp ³	Bent/V-shaped	104.5 deg	H ₂ O, H ₂ S (approx)
sp ³ d	Trigonal Bipyramidal	90/120 deg	PCl ₅
sp ³ d	See-saw	<90/<120	SF ₄
sp ³ d	T-shaped	<90 deg	ClF ₃ , BrF ₃
sp ³ d	Linear	180 deg	XeF ₂ , I ₃ ⁻
sp ³ d ²	Octahedral	90 deg	SF ₆ , PCl ₆ ⁻
sp ³ d ²	Square Pyramidal	<90 deg	IF ₅ , BrF ₅
sp ³ d ²	Square Planar	90 deg	XeF ₄ , [Ni(CN) ₄] ²⁻
sp ³ d ³	Pentagonal Bipyramidal	72/90 deg	IF ₇

2. WORKED EXAMPLES

Example 1 — Hybridisation of XeF₄

Question: Determine the hybridisation and shape of XeF₄.

Solution:

- Step 1: Central atom = Xe; Valence electrons of Xe = 8
- Step 2: Monovalent F atoms = 4; No charge (M = 4, C = 0, A = 0)
- Step 3: $H = (V + M - C + A)/2 = (8 + 4 - 0 + 0)/2 = 12/2 = 6$
- Step 4: $H = 6 \rightarrow sp^3d^2$ hybridisation
- Step 5: 6 hybrid orbitals: 4 used by F (bond pairs) + 2 lone pairs
- Step 6: Lone pairs occupy axial positions (trans to each other) $\rightarrow$ Square Planar geometry
- Answer: sp^3d^2 hybridisation; Square Planar shape; bond angle 90°

Example 2 — Hybridisation of ClF₃

Question: Determine the hybridisation and shape of ClF₃.

Solution:

- Step 1: Central atom = Cl; Valence electrons = 7
- Step 2: M = 3 monovalent F atoms; C = 0, A = 0
- Step 3: $H = (7 + 3)/2 = 10/2 = 5$
- Step 4: $H = 5 \rightarrow sp^3d$ hybridisation
- Step 5: 5 hybrid orbitals: 3 bond pairs + 2 lone pairs
- Step 6: Lone pairs occupy equatorial positions $\rightarrow$ T-shaped geometry
- Answer: sp^3d hybridisation; T-shaped; F-Cl-F axial angle slightly $< 180^\circ$

Example 3 — Hybridisation of SO₂

Question: Determine the hybridisation of S in SO₂.

Solution:

- Step 1: Central atom = S; Valence electrons = 6
- Step 2: SO₂ has resonance; effectively 2 oxygen atoms attached + 1 lone pair
- Step 3: $H = (6 + 0)/2 = 3$ (counting electron domains: 1 double bond + 1 double bond + 1 lone pair = 3 regions)
- Step 4: $H = 3 \rightarrow sp^2$ hybridisation
- Step 5: 3 sp^2 orbitals: 2 used in S=O bonds + 1 for lone pair
- Step 6: Bent/V-shaped geometry; bond angle $\sim 119^\circ$ (slightly less than 120° due to lone pair)
- Answer: sp^2 hybridisation; Bent shape

Example 4 — Hybridisation of NO₂

Question: Determine the hybridisation of N in NO₂.

Solution:

- Step 1: Central atom = N; Valence electrons = 5
- Step 2: $M = 0$ (oxygen not monovalent in standard counting); 2 oxygen atoms
- Step 3: Electron domains on N: 1 double bond + 1 single bond/lone pair + 1 unpaired electron = 3 total
- Step 4: $H = 3 \rightarrow sp^2$ hybridisation
- Step 5: Bond angle $\sim 134^\circ$ (greater than 120° because unpaired electron repels less than a lone pair)
- Answer: sp^2 hybridisation; Bent shape with bond angle $\sim 134^\circ$

Example 5 — Hybridisation of I₃⁻

Question: Determine the hybridisation and shape of I₃⁻ (triiodide ion).

Solution:

- Step 1: Central atom = I; Valence electrons = 7; charge = -1
- Step 2: Total valence electrons with charge: 7 + 2 (from 2 terminal I) + 1 (negative charge) = 10 electrons on central I
- Step 3: $H = (V + M + A)/2 = (7 + 2 + 1)/2 = 10/2 = 5$
- Step 4: $H = 5 \rightarrow sp^3d$ hybridisation
- Step 5: 5 orbitals: 2 bond pairs + 3 lone pairs
- Step 6: 3 lone pairs occupy equatorial positions $\rightarrow$ Linear geometry
- Answer: sp^3d hybridisation; Linear shape; I-I-I angle = 180°

3. KEY FACTS — QUICK REVISION

KEY FACTS / QUICK REVISION

- $H = (V + M - C + A)/2$ gives the hybridisation number directly.
- sp: 2 hybrid orbitals, 180 deg, linear (BeCl₂, CO₂, HCN, C₂H₂)
- sp²: 3 hybrid orbitals, 120 deg, trigonal planar (BF₃, SO₃, C₂H₄)
- sp³: 4 hybrid orbitals, 109.5 deg tetrahedral; lone pairs reduce angle
- sp³d: 5 hybrid orbitals; lone pairs prefer equatorial positions
- sp³d²: 6 hybrid orbitals; lone pairs prefer trans (opposite) positions
- sp³d³: 7 hybrid orbitals (IF₇ only common example)
- Hybridisation describes electron geometry; shape = geometry minus lone pairs
- Bond angle order: sp > sp² > sp³ > sp³d (equatorial) > sp³d²
- XeF₄: sp³d², square planar | XeF₂: sp³d, linear | XeF₆: sp³d³, distorted

4. PRACTICE QUESTIONS — JEE ADVANCED LEVEL

1. The hybridisation of the central atom in XeOF_4 is:

- (A) sp^3d
- (B) sp^3d^2
- (C) sp^3d^3
- (D) sp^2d^2

2. Which of the following species has the maximum number of lone pairs on the central atom?

- (A) ClF_3
- (B) XeF_2
- (C) SF_4
- (D) PCl_5

3. The bond angle in H_2O is less than that in NH_3 because:

- (A) O is more electronegative than N
- (B) H_2O has two lone pairs while NH_3 has one
- (C) O-H bond is longer than N-H bond
- (D) H_2O has sp hybridisation

4. How many sigma bonds are present in one molecule of PCl_5 ?

- (A) 3
- (B) 4
- (C) 5
- (D) 6

5. The geometry of SF₄ is best described as:

- (A) Tetrahedral
- (B) Trigonal bipyramidal
- (C) See-saw/irregular tetrahedron
- (D) Square pyramidal

6. Which of the following has sp³d² hybridisation?

- (A) PCl₅
- (B) SF₄
- (C) XeF₄
- (D) ClF₃

7. In which of the following molecules does the central atom use sp hybridisation?

- (A) SO₂
- (B) CO₂
- (C) H₂O
- (D) BF₃

8. The VSEPR notation for ClF₃ is AX₃E₂. Its hybridisation is:

- (A) sp³
- (B) sp³d
- (C) sp³d²
- (D) sp²d

9. Which set of molecules all have sp^3 hybridisation at the central atom?

(A) NH_3 , H_2O , CCl_4

(B) BF_3 , NH_3 , PCl_3

(C) H_2O , SO_2 , CO_2

(D) CH_4 , C_2H_4 , C_2H_2

10. IF_7 has pentagonal bipyramidal geometry. The hybridisation of I in IF_7 is:

(A) sp^3d^2

(B) sp^3d^3

(C) sp^3d

(D) d^4sp^3

5. ANSWER KEY WITH EXPLANATIONS

1. Answer: B — sp^3d^2

$XeOF_4$: Xe has 8 valence electrons. Bonds: 4 F + 1 O (double bond counts as one domain). Total electron domains = 4 + 1 (double bond) + 1 lone pair = 6. So sp^3d^2 . Shape: Square pyramidal (1 lone pair + 5 bonds).

2. Answer: B — XeF_2

XeF_2 : sp^3d , 2 bond pairs + 3 lone pairs. ClF_3 : 3 bp + 2 lp. SF_4 : 4 bp + 1 lp. XeF_2 has maximum 3 lone pairs.

3. Answer: B

H_2O has 2 lone pairs creating stronger lp-lp and lp-bp repulsion, compressing the bond angle to 104.5 deg. NH_3 has only 1 lone pair giving 107 deg.

4. Answer: C — 5 sigma bonds

PCl_5 has 5 P-Cl bonds, each is a sigma bond. No pi bonds in PCl_5 .

5. Answer: C — See-saw

SF_4 : S has 6 valence electrons, 4 F attached. $H = (6+4)/2 = 5 \rightarrow sp^3d$. 1 lone pair occupies equatorial position. Geometry = see-saw (also called irregular tetrahedron).

6. Answer: C — XeF_4

XeF_4 : $H = (8+4)/2 = 6 \rightarrow sp^3d^2$. XeF_4 has 2 lone pairs in axial positions $\rightarrow$ square planar geometry.

7. Answer: B — CO_2

CO_2 : $O=C=O$. C has 2 double bonds = 2 electron domains $\rightarrow$ sp hybridisation. SO_2 has 3 electron domains (sp^2). H_2O has 4 (sp^3). BF_3 has 3 (sp^2).

8. Answer: B — sp^3d

AX_3E_2 means 3 bond pairs + 2 lone pairs = 5 electron domains total $\rightarrow$ sp^3d hybridisation.

9. Answer: A

NH₃ (4 e- domains, sp³), H₂O (4 e- domains, sp³), CCl₄ (4 e- domains, sp³) — all sp³. BF₃ and SO₂ are sp²; C₂H₄ and CO₂ involve sp² and sp respectively.

10. Answer: B — sp³d³

IF₇: I has 7 valence electrons + 7 F atoms (monovalent). $H = (7+7)/2 = 7 \rightarrow \text{sp}^3\text{d}^3$. 5 equatorial + 2 axial positions give pentagonal bipyramidal geometry.